

Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values

by

Thabang Bongani Junior Baloyi

Dissertation submitted in partial fulfilment of the requirements in respect
of the degree

Master's Degree

(Mathematical Statistics & Actuarial Sciences)

in the Department of Mathematical Statistics & Actuarial Science
in the Faculty of Natural and Agricultural Sciences
at the University of the Free State



UNIVERSITY OF THE FREE STATE
UNIVERSITEIT VAN DIE VRYSTAAT
YUNIVESITHI YA FREISTATA

Bloemfontein Campus, South Africa
May 2026

Supervisor: Dr JM Blomerus